Project Title: EvacRoute - An Intelligent Evacuation Route Planner

---

1. Introduction: Natural disasters and emergencies such as earthquakes, floods, fires, and industrial accidents pose significant threats to human life. Rapid and efficient evacuation planning plays a critical role in minimizing casualties. EvacRoute is an intelligent evacuation route planner designed to assist individuals in navigating to the safest exit during such emergencies by using real-time map data and the A* algorithm for optimal route finding.

2. Objective: The primary objective of this project is to develop a system that can:

- Display a map of the affected area.
- Identify and mark danger zones.
- Calculate and display the safest and shortest evacuation route using the A* (A-star) algorithm.

3. Tools and Technologies Used:

- **Programming Language:** Python
- **Algorithm:** A* (A-star) algorithm
- **Map Integration:** Folium and OpenStreetMap
- **Frontend:**Streamlit

4. Working Principle:

- Users input their current location via the web interface.
- The map displays the surrounding area, including exits and marked danger zones.
- The A* algorithm calculates the optimal path to the nearest safe zone considering obstacles and hazards.

- The route is highlighted on the map.

5. A Algorithm Overview:** The A* algorithm is a pathfinding and graph traversal algorithm known for its performance and accuracy. It uses a heuristic to evaluate nodes and find the most efficient path from a start node to a goal node.

- **f(n) = g(n) + h(n)**
    - *g(n)*: Actual cost from the start node to the current node.
    - *h(n)*: Estimated cost from the current node to the goal.
- In EvacRoute, the nodes are positions on the map grid, and the algorithm avoids danger zones (marked as high-cost or impassable).

6. Features:

- Interactive user interface for location input.
- Real-time route planning.
- Visual display of danger zones and safe paths.
- Optimized path generation using A* algorithm.
- Scalable design for integration with real-world GIS data and emergency systems.

7. Applications:

- Urban disaster management
- Emergency response planning
- Evacuation drills and simulations
- Public safety and smart city integration

8. Future Enhancements:

- Integration with live disaster alert systems.

- Mobile application development.
- AI-based prediction of danger zones using historical data.
- Multi-user routing with congestion prediction.

9. Conclusion: EvacRoute presents a smart, responsive solution to aid in emergency evacuations. By leveraging the efficiency of the A* algorithm and real-time mapping technologies, it provides a reliable tool for safe and fast evacuation during critical situations. This project not only enhances technical skills in algorithms and web development but also addresses a vital societal need.
